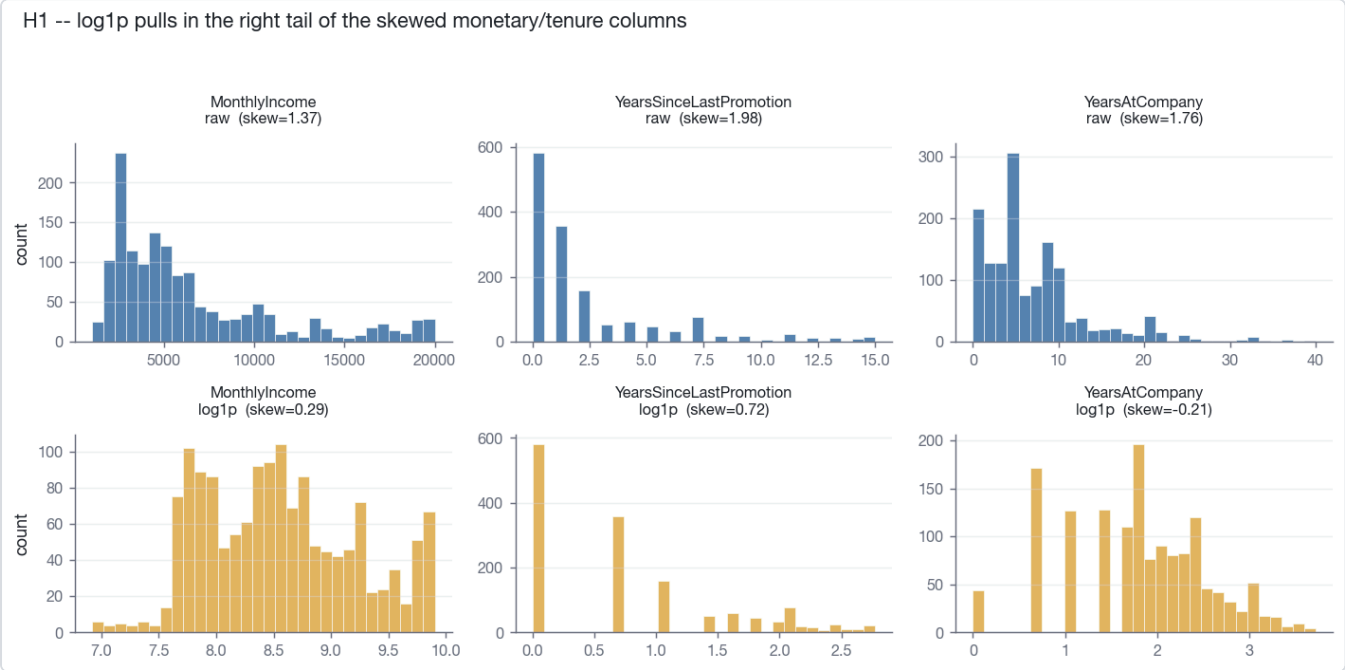
Assignment #3 — Step 3: Hypotheses

Every hypothesis below follows the required chain **EDA finding** → **proposed change** → **expected effect**, sourced from the EDA in `notebooks/00_baseline.ipynb` (Part 1) and from the targeted figures generated for this step (`src/hypotheses_figures.py` → `outputs/figures/08_*.png` – `12_*.png`). Baseline (E0) to beat: **LR + max-out, F1 = 0.464 (95% CI 0.31–0.60)** — see `README.md`.

All proposed changes are fit inside the training fold only (no leakage) and change **one pipeline component at a time**, per the brief's E0/E1/E2/E3 protocol. None of them replace the main model — logistic regression stays the model under test throughout, exactly as Assignment #3 requires.

H1 — Log-transform the skewed monetary/tenure columns

EDA finding. `MonthlyIncome`, `YearsSinceLastPromotion`, and `YearsAtCompany` are strongly right-skewed (skew 1.37, 1.98, 1.76; 7.1–7.8% of values sit beyond the 1.5×IQR fence — `eda_outliers.csv`).



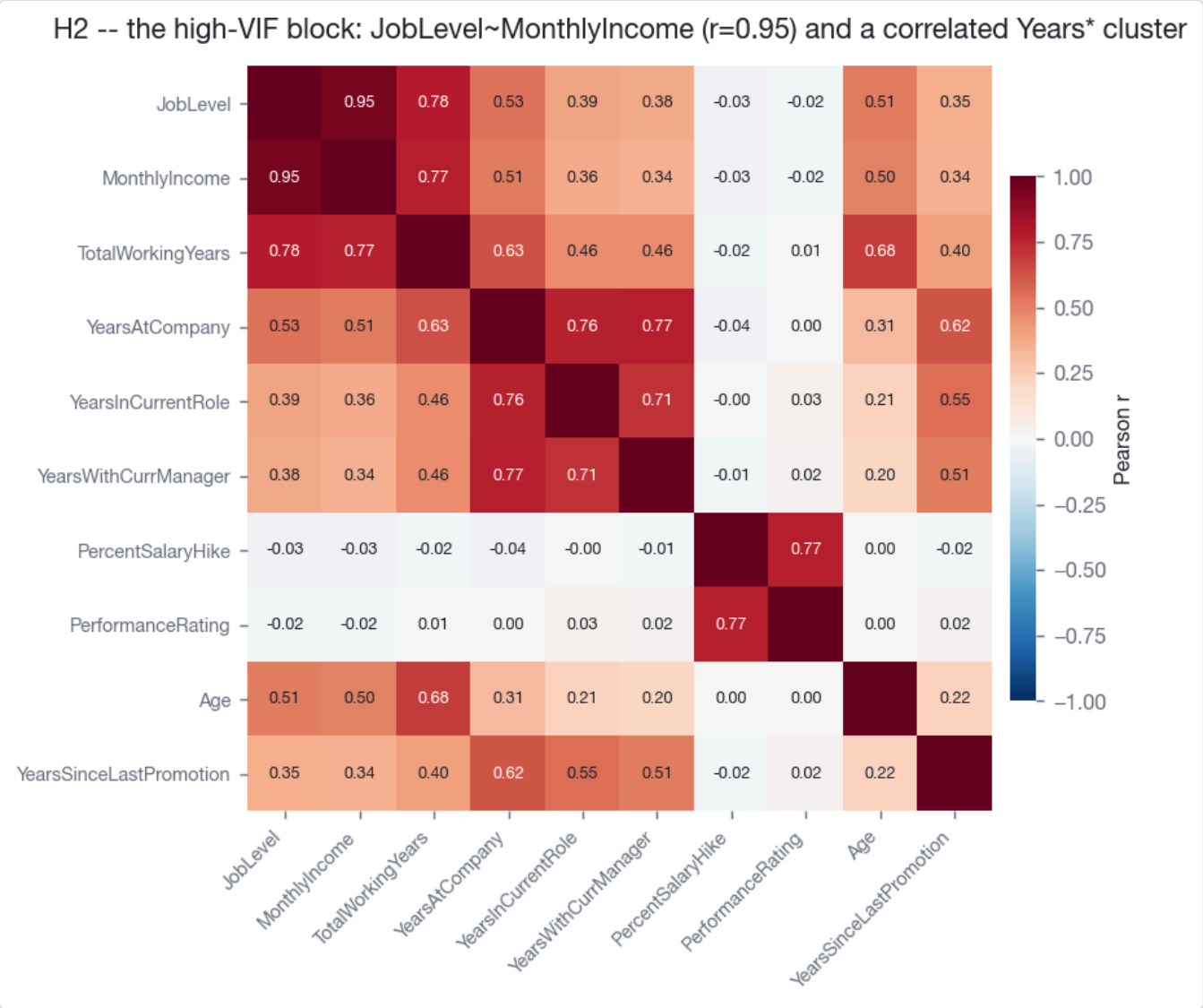
A `log1p` transform pulls skew down to 0.29, 0.72, and −0.21 respectively — close to symmetric.

Proposed change. Apply `log1p` to these three columns (and any other column with |skew| > 1) inside the `ColumnTransformer`, fit on training rows only, before standardisation.

Expected effect. Logistic regression is a linear-in-the-features model; a few extreme incomes/tenures currently exert disproportionate leverage on the fitted boundary and on `StandardScaler`'s mean/SD. Compressing the tail should produce more stable coefficients (lower bootstrap CV, Part 2.6 of the baseline notebook) and may modestly improve F1 and ROC-AUC by making the decision boundary less distorted by a handful of senior, long-tenured employees who are overwhelmingly non-leavers.

H2 — Resolve the JobLevel / MonthlyIncome redundancy

EDA finding. `JobLevel` and `MonthlyIncome` correlate at r = 0.95 (VIF 11.2 and 10.8 — the two highest in the dataset); `TotalWorkingYears` and the four `Years*` tenure columns form a second correlated block (r = 0.51–0.77).

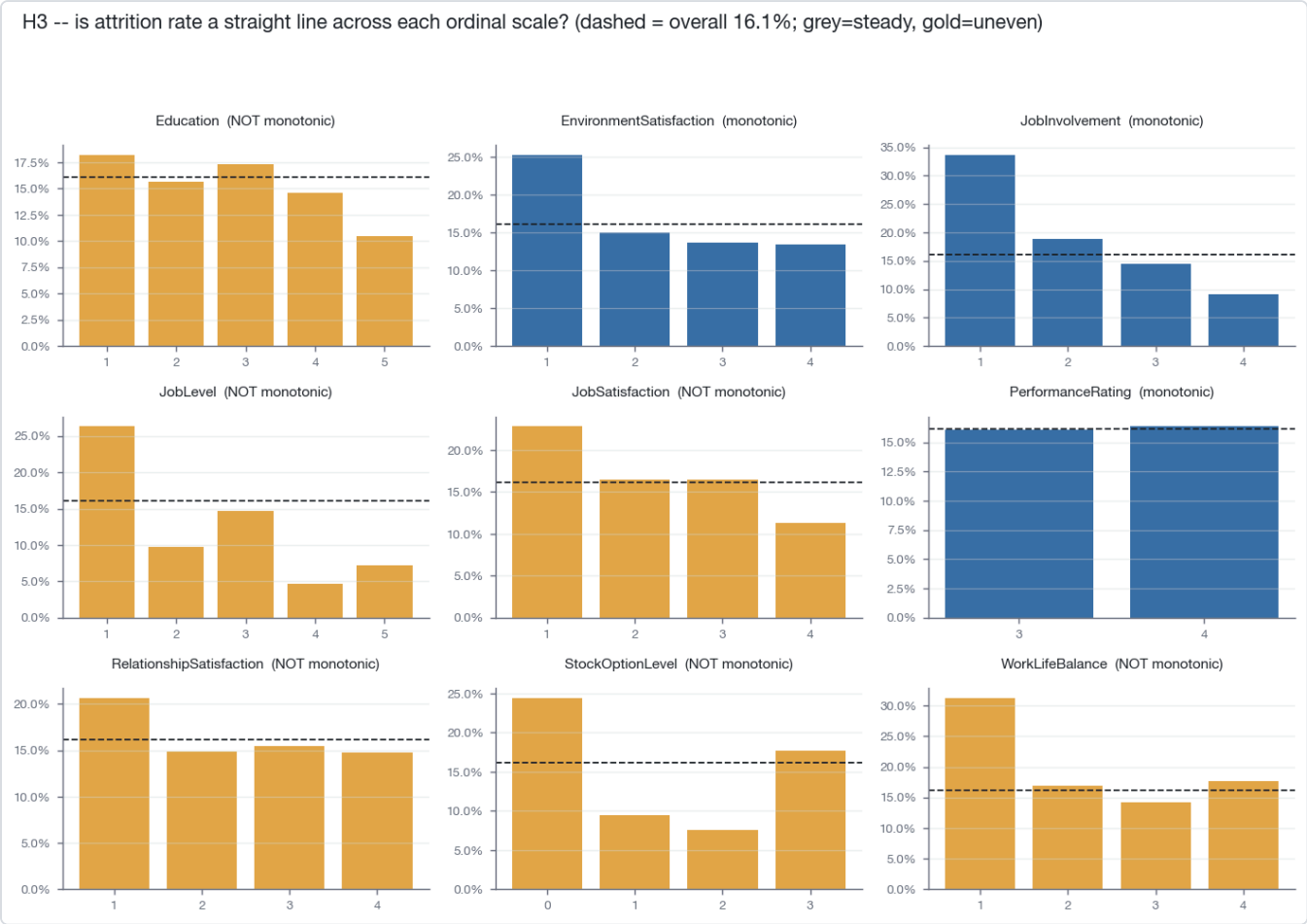


Proposed change. Drop `JobLevel` and keep `MonthlyIncome` (income is the finer-grained of the near-duplicate pair and is already the stronger Cohen's-d signal per `eda_numeric_association.csv`), rather than relying on max-out's wrapper search alone to make that call.

Expected effect. Removing a near-duplicate feature should shrink the coefficient instability the bootstrap analysis already flags for this block (Part 2.6, baseline notebook) without costing predictive signal, since `JobLevel`'s information is ~95% contained in `MonthlyIncome`. A null result (no F1 change) is also plausible and would itself be informative: max-out already tends to discard the same redundant dummies it can reach, so an explicit pre-filter may only replicate work the wrapper already does.

H3 — Stop treating the ordinal survey scales as equal-interval numbers

EDA finding. Nine 1–4/1–5 survey scales (`Education` , `EnvironmentSatisfaction` , `JobInvolvement` , `JobLevel` , `JobSatisfaction` , `PerformanceRating` , `RelationshipSatisfaction` , `StockOptionLevel` , `WorkLifeBalance`) are fed to the model as continuous numbers, which assumes attrition rate moves in a straight line across levels.



Checking that assumption directly: **6 of 9** scales are *not* monotonic. `Education` and `JobSatisfaction` dip and recover; `JobLevel` , `RelationshipSatisfaction` , `StockOptionLevel` , and `WorkLifeBalance` show level-1 (or level-0) employees as outliers rather than one end of a trend. Only `EnvironmentSatisfaction` , `JobInvolvement` , and the near-binary `PerformanceRating` are genuinely monotonic.

Proposed change. One-hot encode the six non-monotonic scales instead of passing them through as numeric; leave `EnvironmentSatisfaction` , `JobInvolvement` , and `PerformanceRating` numeric, since a straight line is already a reasonable fit for those three.

Expected effect. For the six non-monotonic scales, one-hot encoding lets logistic regression fit a separate coefficient per level instead of forcing a single slope through a non-linear relationship, which should improve fit on those features specifically. The cost is more parameters (helpful context for how aggressively max-out needs to prune afterward) and a real risk that gains are undone by the accompanying variance increase on a 221-row test set — a case where a "no help" result is plausible and worth reporting either way.

H4 — Compare class-weighting against fold-safe oversampling

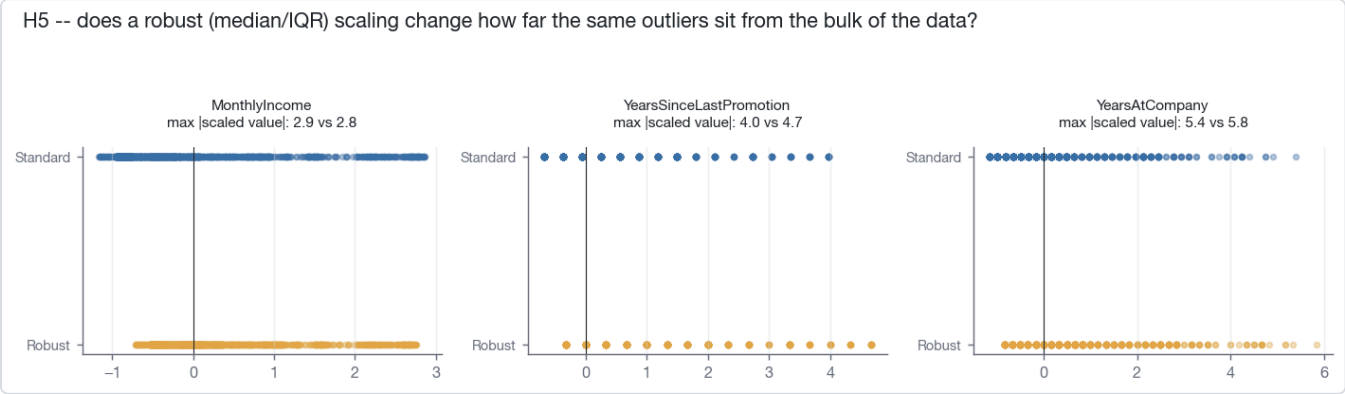
EDA finding. The target is 16.1% positive (237 of 1470 — `eda_class_counts.csv`, figure `01_class_balance.png` in the baseline notebook). The baseline already applies `class_weight="balanced"` to correct for this in the loss function.

Proposed change. As an alternative to (not necessarily a replacement for) class weighting, fit `SMOTE` / `RandomOverSampler` **inside each training fold only** (never on validation/test) and compare against the `class_weight="balanced"` baseline at the same validation-tuned threshold.

Expected effect. Oversampling and class weighting attack the same problem through different mechanisms (synthetic minority examples that inflate density near the decision boundary, vs. a reweighted loss). Oversampling could give logistic regression a smoother minority-class boundary, but could equally add synthetic-sample noise; on a class imbalance this mild (5.2:1), class weighting alone may already capture most of the achievable benefit. A result of "no significant difference" is a legitimate and useful outcome here.

H5 — Use a robust (median/IQR) scaler on the skewed columns

EDA finding. The same three skewed columns from H1 have real outliers by value (max `MonthlyIncome` = 19,999 against an upper fence of 16,581), not just by shape.



Checking the actual scaled magnitudes overturned the naive assumption: `RobustScaler` does **not** shrink these outliers relative to `StandardScaler` — for `YearsSinceLastPromotion` (max |z| 4.0 → 4.7 under robust scaling) and `YearsAtCompany` (5.4 → 5.8) the outliers land *further* out under the robust scaling, and `MonthlyIncome` is essentially unchanged (2.9 vs 2.8). The reason is **outlier self-masking**: an outlier inflates the very standard deviation used to standardise it, which shrinks its own z-score under `StandardScaler`. A median/IQR scale isn't computed from the outliers' own spread, so it doesn't mask them the same way — it reports how far a point sits from the *typical* employee, without letting a few extreme employees widen the ruler.

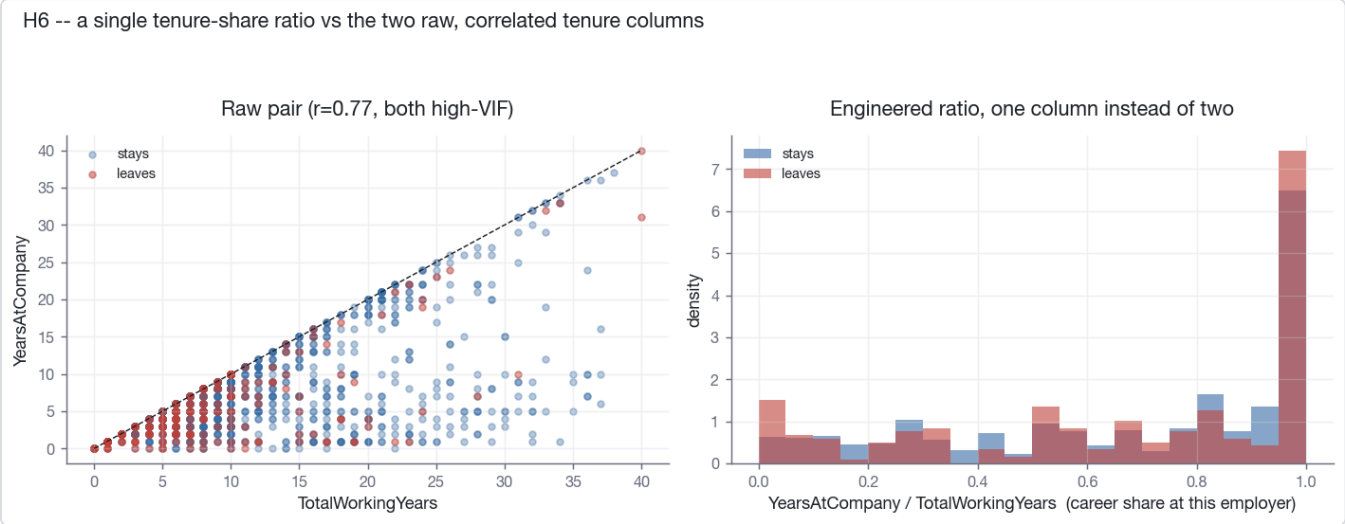
Proposed change. Replace `StandardScaler` with `RobustScaler` for the skewed numeric block (test this both with and without the H1 log transform, since the two address the same symptom through different, and now known to be non-equivalent, mechanisms).

Expected effect. Because `RobustScaler` reports these outliers as *more* extreme rather than less, the direction of any performance effect is genuinely uncertain and this is exactly why it needs an experiment rather than a theoretical argument: a scale that doesn't self-mask outliers could help logistic regression treat them as the meaningfully distant points they are (better separation for genuinely unusual employees), or it

could increase their leverage on the fitted boundary and hurt generalisation on the 221-row test set. A result showing `RobustScaler` performs about the same as, or worse than, `StandardScaler` here would be a valid and informative outcome, not a failed experiment.

H6 — Replace the two raw tenure columns with one engineered ratio

EDA finding. `TotalWorkingYears` and `YearsAtCompany` are both high-signal (top Cohen's-d features) *and* highly collinear (r = 0.77, VIF 4.8 and 4.6) — the same redundancy problem as H2, but for a pair that is harder to simply drop one side of, since both carry independent information (career length vs. loyalty to this employer).



Proposed change. Engineer a single `tenure_share = YearsAtCompany / TotalWorkingYears` feature (career share spent at the current employer, clipped to [0, 1]) and let max-out choose between keeping this ratio, keeping the two raw columns, or keeping both.

Expected effect. A compact ratio that captures the same relative information as two correlated raw columns should let the max-out search eliminate more redundant features without losing predictive signal, and should further stabilise the coefficient-bootstrap results for this block (same mechanism as H2). The density plot above shows a real but modest separation (leavers cluster slightly more at low `tenure_share` — i.e. newer to the company relative to their total career — while stayers show a second mode near 1.0), so the expected F1 effect is small; the more likely win is a leaner, more stable model rather than a large accuracy gain.

H7 — Pre-filter by correlation/VIF before running max-out

EDA finding. The baseline's leakage-safe max-out search already eliminates 8 LR features (`notebooks/00_baseline.ipynb`, Part 2.3), and every one of them sits inside the correlated/high-VIF blocks identified in H2 and section 1.4 of the EDA (`Department_Research & Development`, `Education`, `EducationField_Human Resources`, `EducationField_Marketing`, `Gender_Female`, `JobLevel`, `MonthlyRate`, `PerformanceRating`). Max-out is already, implicitly, doing correlation-driven pruning — just slowly, via 1,450+ greedy refits across 7 random splits.

Proposed change. Add an explicit pre-filter — drop one feature from any pair with $|r| \geq 0.85$ or VIF ≥ 10, chosen by lower target association — fit on the training fold, *before* handing the reduced feature set to max-out.

Expected effect. Since the greedy search already converges on removing most of these features, the pre-filter is expected to produce a similar final feature set and similar F1, but with less run-to-run variance across the 7 max-out resamples (fewer near-duplicate features competing to be the one dropped first) and a substantially faster search. If F1 does not improve, that is still evidence worth reporting: it would show the wrapper search was already finding the efficient solution on its own, making the extra pre-filtering step unnecessary complexity rather than a genuine improvement.

Summary — hypotheses at a glance

#	EDA finding	Proposed change	Primary expected effect	Addresses
H1	Skew 1.37–1.98 in 3 monetary/tenure cols	`log1p` transform (train-fit)	More stable LR coefficients; possible small F1/AUC gain	Numerical transformation
H2	`JobLevel` ~ `MonthlyIncome` r=0.95, VIF 11	Drop `JobLevel`	Stabler coefficients, F1 roughly unchanged	Redundant-feature removal
H3	6/9 ordinal scales non-monotonic vs attrition	One-hot the 6 non-monotonic scales	Better fit on those features; may be offset by variance	Categorical encoding
H4	16.1% positive class	Fold-safe SMOTE vs `class_weight`	Uncertain — legitimate to find no difference	Missing/imbalance handling
H5	Same skewed cols have real-value outliers; `StandardScaler` self-masks them (robust scaling shows them as *more*, not less, extreme)	`RobustScaler` (median/IQR)	Direction genuinely uncertain — needs the experiment	Scaling/normalisation
H6	`TotalWorkingYears` ~ `YearsAtCompany` r=0.77, both high-signal	Engineer `tenure_share` ratio	Leaner model,	Feature engineering

#	EDA finding	Proposed change	Primary expected effect	Addresses
			small/no F1 change	
H7	Max-out already removes exactly the correlated/high-VIF features	Explicit correlation/VIF pre-filter before max-out	Similar F1, lower variance, faster search	Feature selection

Experiment mapping (Step 4 preview)

Each hypothesis becomes its own experiment against the shared E0 baseline, changing one component at a time, evaluated on identical stratified k-fold CV splits:

Experiment	Baseline +	Tests
E1	H1 (log1p transform)	H1 alone
E2	H5 (RobustScaler)	H5 alone, and vs. E1
E3	H1 + H5 combined	whether the two skew/outlier fixes are additive or redundant
E4	H2 (drop `JobLevel`)	H2 alone
E5	H3 (one-hot non-monotonic ordinals)	H3 alone
E6	H6 (`tenure_share` ratio)	H6 alone
E7	H4 (fold-safe SMOTE, in place of `class_weight`)	H4 alone
E8	H7 (correlation/VIF pre-filter + max-out)	H7 alone
E9	best-performing subset from E1–E8	selected combination

Hyperparameter tuning (Step 5) and the final test-set evaluation (Step 6) happen only after this grid is complete, on top of whichever preprocessing combination E9 selects.